

ZAPATA COUNTY AP, TX, USA

WMO: 720584

Lat: 26.967N

Lon: 99.250W

Elev: 129

StdP: 99.78

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00181

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	2.4	-10.2	1.6	13.6	-7.8	2.0	12.2	10.0	19.4	8.9	17.5	1.2	350	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	40.2	22.1	39.2	22.4	38.1	22.5	25.6	30.8	25.2	30.2	24.9	30.1	4.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.0	26.3	24.1	19.3	26.4	23.7	18.8	26.5	78.9	29.7	77.5	30.0	76.0	29.9	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	8.0	7.3	-3.7	42.1	2.4	0.8	-5.5	42.6	-6.9	43.1	-8.2	43.5	-10.0	44.0		
			-4.6	26.2	2.1	0.7	-6.1	26.7	-7.4	27.0	-8.5	27.4	-10.1	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.8	13.9	17.4	21.0	24.9	27.9	30.6	31.4	31.5	28.4	24.2	18.8	14.9
	DBStd	7.35	5.06	5.76	4.97	3.83	3.09	1.90	1.63	1.66	2.62	4.38	5.30	5.02
	HDD10.0	46	20	9	2	0	0	0	0	0	0	0	4	12
	HDD18.3	470	154	81	34	4	0	0	0	0	0	9	58	131
	CDD10.0	5073	142	216	343	446	554	617	663	667	553	439	268	165
	CDD18.3	2456	17	54	118	200	296	367	404	409	303	190	72	26
	CDH23.3	29788	162	469	1046	2087	3428	4898	5576	5741	3502	2045	640	196
	CDH26.7	16064	36	167	434	1038	1754	2831	3329	3499	1818	910	204	42
Wind	WSAvg	3.3	2.5	3.0	3.5	3.9	4.1	4.1	4.2	3.7	2.6	2.7	2.5	2.3
Precipitation	PrecAvg	542	23	22	22	31	59	52	61	43	107	42	38	20
	PrecMax	847	96	76	66	103	152	156	174	98	262	148	99	85
	PrecMin	337	0	0	1	1	0	2	1	1	23	2	0	1
	PrecStd	137	27	22	20	31	44	37	54	32	56	36	30	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	34.0	35.7	39.7	39.0	41.5	41.1	41.3	39.0	36.7	32.8	30.3
		MCWB	17.4	17.9	18.3	19.6	22.7	22.6	21.8	22.1	21.9	23.0	21.2	18.6
	2%	DB	27.4	30.9	32.8	36.3	37.5	39.2	40.0	40.1	37.5	34.6	31.1	27.7
		MCWB	16.9	18.3	19.1	20.0	22.9	22.9	22.2	22.2	22.1	22.4	20.6	18.3
	5%	DB	25.0	28.6	31.2	34.1	36.0	37.9	39.0	39.1	36.2	32.9	29.1	25.5
		MCWB	15.7	18.5	19.3	20.5	22.7	22.8	22.5	22.3	22.3	22.0	20.0	17.6
	10%	DB	22.5	26.3	29.0	32.4	34.1	36.9	37.8	38.0	34.8	31.8	27.2	23.1
		MCWB	15.5	18.1	19.0	20.1	22.7	22.7	22.6	22.5	22.4	21.8	19.0	17.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	21.7	22.3	24.1	25.7	25.8	25.8	25.7	25.4	23.8	21.6	
		MCDB	24.8	27.2	27.1	30.2	31.4	32.6	30.9	30.5	30.4	30.0	27.6	24.8
	2%	WB	19.5	20.8	21.8	23.2	25.0	25.2	25.3	25.3	25.2	24.8	22.9	20.7
		MCDB	23.9	25.4	26.9	29.6	30.8	30.9	30.4	30.1	29.1	29.1	26.8	24.3
	5%	WB	18.5	20.3	21.2	22.6	24.5	24.9	24.8	24.9	24.8	24.2	21.8	19.9
		MCDB	22.0	24.7	26.3	29.0	30.6	30.7	30.2	30.3	29.0	28.6	26.0	23.3
	10%	WB	17.2	19.7	20.6	22.0	24.0	24.5	24.4	24.5	24.5	23.6	20.9	18.8
		MCDB	20.6	24.0	25.6	28.3	30.3	30.5	30.0	30.4	28.8	28.4	25.5	22.3
Mean Daily Temperature Range	5% DB	MDBR	13.1	12.9	12.7	13.8	12.1	12.6	12.9	13.3	11.4	12.6	12.4	11.8
		MCDBR	16.0	15.4	14.7	15.5	13.7	14.0	14.4	14.6	13.5	12.5	13.2	14.6
		MCWBR	7.6	5.4	4.9	4.6	2.9	2.6	2.6	2.7	3.0	3.4	4.9	7.0
	5% WB	MCDBR	11.5	11.5	12.3	13.2	11.4	11.9	12.1	12.2	10.3	10.5	10.7	11.1
		MCWBR	6.4	5.0	4.8	4.6	3.0	2.8	2.6	2.7	2.6	3.6	4.5	6.4
Clear-Sky Solar Irradiance	taub	0.326	0.347	0.374	0.398	0.413	0.406	0.399	0.398	0.397	0.364	0.355	0.329	
	taud	2.524	2.461	2.394	2.347	2.347	2.424	2.439	2.465	2.426	2.509	2.472	2.522	
	Ebn at Noon	926	929	918	901	882	881	887	890	884	899	884	906	
	Edh at Noon	91	104	118	127	127	117	115	112	113	98	95	88	
All-Sky Solar Radiation	RadAvg	3.22	3.96	4.88	5.68	6.05	6.89	6.82	6.60	5.36	4.59	3.57	2.95	
	RadStd	0.39	0.66	0.62	0.55	0.51	0.31	0.50	0.31	0.45	0.52	0.42	0.45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page